

Every body perseveres in a state of rest, or of uniform motion in a right line, unless compelled to change that state. A body left to itself does nothing on its own — it neither slows, speeds, nor turns. The sole agent that alters this perseverance is a force impress'd thereon, so that with $\sum \vec{F} = 0$ we have $\vec{v} = \text{const}$: velocity, in both speed and direction, is preserved.